

The New York City Department of Education

2024 Specialized High Schools Admissions Test**GENERAL DIRECTIONS****Student Name:** _____**Identifying Information**

Turn to Side 1 of the answer sheet.

Notify the proctor immediately if you are ill or should not be taking this test. Do not sign the statement or begin the test. Return your answer sheet to the proctor.**Line 1:** Read the statement and sign your name in the space following the word “signature.” Do not print your name.**Line 2:** Print today’s date, using the numbers of the month, the day, and the year. For example, the date September 21, 2024, would be 9-21-24.**Line 3:** Print your birth date, using the numbers of the month, the day, and the year. For example, a birth date of March 1, 2009, would be 3-1-09.**Grid 4:** Print the letters of your first name, or as many as will fit, in the boxes. If you have a middle initial, print it in the box labeled “MI.” Then print the letters of your last name, or as much as will fit, in the boxes provided. Below each box, fill in the circle that contains the same letter as the box. If there is a space or a hyphen in your name, fill in the circle under the appropriate blank or hyphen.**Make dark marks that completely fill the circles.** If you change a mark, be sure to erase the first mark completely.**Grid 5:**

1. Print the name of the school where you are now enrolled in the space at the top of the grid.
2. In the boxes marked “SCHOOL CODE,” print the six-digit code that identifies your school. Below each box, fill in the circle that contains the same letter or number as the box. (You can find your school code on your Test Ticket. If it is not there, tell the proctor, and the proctor will get the school code for you.)
3. If you attend a private or parochial school, fill in the circle marked “P.”

Grid 6: Complete the grid with your date of birth. Print the first three letters of the month in the first box, the number of the day in the next box, and the year in the last box. Then fill in the corresponding circles.**Grid 7:** Print your student ID number in Grid 7. You can find your student ID number on your Test Ticket. In the boxes, print your nine-digit student ID number. Below each box, fill in the circle containing the same number as in the box.**Grid 8:** In most cases, Grid 8 is already filled in for you. If it is not, copy the letter and numbers shown in the upper-right corner of your test booklet into the boxes. Below each box, fill in the circle containing the same letter or number as the box.

Now review Side 1 to make sure you have completed all lines and grids correctly. Review each column to see that the filled-in circles correspond to the letters or numbers in the boxes above them.

Turn your answer sheet to Side 2 and flip this booklet over to continue with General Directions.

**DO NOT OPEN THIS BOOKLET
UNTIL YOU ARE TOLD TO DO SO.****TURN YOUR BOOKLET OVER TO THE BACK COVER.**

GENERAL DIRECTIONS, continued

Print your test booklet letter and numbers, and your name, first name **first**, in the spaces provided at the top of Side 2 of the answer sheet.

Marking Your Answers

- Mark each of your answers on the answer sheet in the row of circles matching the question number printed in the test booklet.
- Use only a Number 2 pencil.
- If you change an answer, be sure to erase it completely.
- Be careful to avoid making any stray pencil marks on your answer sheet. **Do not make any marks on your answer sheet other than filling in your answer choices.**
- Each question has only one correct answer. If you mark more than one circle in any answer row, that question will be scored as incorrect.

SAMPLE ANSWER MARKS				
(A)	(B)	(C)	●	RIGHT
(A)	(B)	(C)	(D)	WRONG
(A)	(B)	(C)	(D)	WRONG
(A)	(B)	(C)	(D)	WRONG
(A)	(B)	●	(D)	WRONG
(A)	(B)	●	●	WRONG

- You can use your test booklet or the provided scrap paper to take notes or solve questions; however, your answers must be recorded on the answer sheet in order to be counted.
- **You will not be able to mark your answers on the answer sheet after time is up, and answers left in the test booklet will not be scored.**

Planning Your Time

- You have 180 minutes to complete the entire test. **How you divide the time between the English Language Arts and Mathematics sections is up to you.**
- If you begin with the English Language Arts section, you may go on to the Mathematics section as soon as you are ready. Likewise, if you begin with the Mathematics section, you may go on to the English Language Arts section as soon as you are ready.
- Be sure to read the directions for each section carefully.
- Each question has only one correct answer. Choose the best answer for each question.
- When you finish a question, go on to the next, until you have completed the last question or run out of time.
- Your score is determined by the number of questions you answer correctly. **Answer every question, even if you may not be certain which answer is correct.**
- Don't spend too much time on a difficult question. Come back to it later if you have time.
- If you complete the test before the test time (180 minutes) is over, you may go back to review questions in either section.
- Students must stay for the entire test session (180 minutes).

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO



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PART 1 — ENGLISH LANGUAGE ARTS

57 QUESTIONS

REVISING/EDITING

QUESTIONS 1-15 (PART A AND PART B)

REVISING/EDITING PART A

DIRECTIONS: Read and answer the following questions. You will be asked to recognize and correct errors so that the sentences or short paragraphs follow the conventions of standard written English. You may write in your test booklet as needed to take notes. You should reread relevant parts of the sentences or paragraphs, while being mindful of time, before marking the **best** answer for each question.

1. What is the **best** way to combine these sentences to clarify the relationship between ideas?

(1) The International Space Station has been inhabited by crew members since 2000.
(2) Tourists will soon be allowed to pay for visits to the space station.
(3) Because the cost is \$52–\$58 million round trip, few people will be able to take advantage of the opportunity to have a vacation in outer space.

- A.** The International Space Station has been inhabited by crew members since 2000, but tourists will soon be allowed to pay for visits to the space station, which cost \$52–\$58 million round trip, so few people will be able to take advantage of the opportunity to have a vacation in outer space.
- B.** The International Space Station has been inhabited by crew members since 2000 and will soon allow tourists to pay for visits, but because the cost is \$52–\$58 million round trip, few people will be able to take advantage of the opportunity to have a vacation in outer space.
- C.** The International Space Station has been inhabited by crew members since 2000, but tourists will soon be allowed to pay for visits to the space station, though the cost is \$52–\$58 million round trip, so few people will be able to take advantage of the opportunity to have a vacation in outer space.
- D.** The International Space Station has been inhabited by crew members since 2000, and tourists will soon be allowed to pay for visits to the space station because the cost is \$52–\$58 million round trip, therefore few people will be able to take advantage of the opportunity to have a vacation in outer space.

2. Which sentence contains an error in its construction and should be revised?

(1) In 1976, the National Basketball Association (NBA) absorbed several teams of the American Basketball Association (ABA), including the New York Nets, who played in the Long Island area at the time. (2) The owner of the Nets decided to take the team to New Jersey after the team had financial troubles, where the team played for thirty-five seasons. (3) The New Jersey Nets had sixteen playoff appearances, including two appearances in the NBA finals. (4) In 2012, the team changed ownership and returned to New York, where the team now plays under the name the Brooklyn Nets.

- E. sentence 1
- F. sentence 2
- G. sentence 3
- H. sentence 4

3. Read this sentence.

The Appalachian Trail is a really long trail that a lot of people do each year.

Which revision of the sentence uses the **most** precise language?

- A. The Appalachian Trail is an extremely long trail that millions of people do each year.
- B. The Appalachian Trail is a 2,200-mile trail that more than a million people hike each year.
- C. The Appalachian Trail is a 2,200-mile trail that two million people hike each year.
- D. The Appalachian Trail is a lengthy trail that a couple million people do each year.

4. Which pair of revisions need to be made in this paragraph?

(1) When coal was used to heat homes, it frequently left soot stains on the walls. (2) Brothers Cleo and Noah McVicker, who owned a cleaning product company created a doughy substance to help people remove this soot. (3) Over time, as natural gas becomes more common, people had little need for soot cleansers, and the McVickers' family company struggled to stay in business. (4) Then one day Joe McVicker, Cleo's son, learned that his sister-in-law had been using the substance for art projects in her classroom, so he remarketed the product as the toy known today as Play-Doh.

- E. Sentence 1: Delete the comma after **homes**.
Sentence 3: Change **becomes** to **became**.
- F. Sentence 1: Delete the comma after **homes**.
Sentence 4: Change **remarketed** to **had remarketed**.
- G. Sentence 2: Insert a comma after **company**.
Sentence 3: Change **becomes** to **became**.
- H. Sentence 2: Insert a comma after **company**.
Sentence 4: Change **remarketed** to **had remarketed**.

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REVISING/EDITING PART B

DIRECTIONS: Read each of the following two texts and answer the related questions. You will be asked to improve the writing quality of each text and to correct errors so that the texts follow the conventions of standard written English. You should reread relevant parts of each text, while being mindful of time, before marking the **best** answer for each question.

Find Time to Volunteer

(1) Many people believe that they gain a greater sense of purpose by giving their time to serve others. (2) These are the people who spend their free time volunteering at various places within the community. (3) Those who are served benefit from the work of these volunteers. (4) However, research shows that those who volunteer also receive benefits.

(5) For many students, this proposition may sound impossible. (6) Schoolwork can require hours of study each week. (7) Volunteering at a retirement center or homeless shelter requires both time and energy. (8) Extracurricular activities fill up after-school time. (9) Family obligations and part-time jobs often have to be worked into the schedule, too. (10) However, if students can find time to volunteer, even for a few hours a month, they may find that the benefits outweigh the scheduling difficulties.

(11) To begin with, students who engage in volunteer activities acquire valuable skills and experience, which can help them focus on potential fields of study and career options.

(12) Furthermore, when students list volunteer work on college applications, admissions counselors see applicants who care about making their community and college campus a better place.

(13) Volunteerism benefits more than students applying to college. (14) This type of experience is equally important for students moving directly into the workforce after high school.

(15) Volunteering demonstrates initiative, dedication, and a strong work ethic, qualities that companies value in a potential employee. (16) In a CareerBuilder study, 60 percent of managers regarded volunteerism as a significant asset when considering applicants.

(17) Students who volunteer undoubtedly make a positive investment in their future, but they also affect their present quality of life. (18) Studies have shown that volunteering is good for both the mind and the body. (19) Volunteering can boost mood levels and reduce anxiety. (20) According to the Mayo Clinic,¹ the feeling of doing something meaningful and the appreciation received from that action can reduce stress.

(21) Even the busiest of students can find a way to donate some time to a worthy cause.

(22) Though managing different commitments may seem overwhelming at times, students are sure to feel a sense of fulfillment in volunteerism.

¹**Mayo Clinic:** a nonprofit medical organization dedicated to treatment and research

5. Which revision of sentence 2 uses the **most** precise language?
- A. These are the people who spend their free time volunteering at animal shelters, helping with activities in community centers, or cleaning up parks.
 - B. These are the people who spend their free time helping others in numerous ways at a variety of places, events, or organizations that need support.
 - C. These are the people who spend their free time working at local establishments that help people, animals, or other groups in need of assistance.
 - D. These are the people who spend a lot of time volunteering at places where they can help people in many ways.
6. Which sentence should follow sentence 4 to introduce the main claim of the passage?
- E. With that in mind, high school students should consider engaging in some form of regular volunteerism.
 - F. Fortunately for students, these benefits are guaranteed to produce both immediate and long-term results.
 - G. In fact, studies have confirmed that volunteerism can be beneficial for students, the family, and the community.
 - H. For this reason, high school students should learn about how helping others can strengthen their communities.
7. Which sentence is least relevant to the ideas presented in the second paragraph (sentences 5–10) and should be deleted?
- A. sentence 6
 - B. sentence 7
 - C. sentence 8
 - D. sentence 9

8. Which of these would **best** follow sentence 12 and support the ideas presented in the third paragraph (sentences 11–12)?
- E. These students show that they are likely to be responsible, reliable, and helpful young adults. Admissions counselors know that character traits like these will benefit the applicants as well as whomever they interact with.
 - F. Volunteering can help students meet new people who can provide references and advice to the students when they are applying to colleges. It can also be an invaluable opportunity for students to meet possible future employers.
 - G. The counselors hope that young adults who demonstrate this behavior will go on to volunteer during their college years, working on campus in student government and leadership programs or serving in the community at hospitals, schools, and homeless shelters.
 - H. Many colleges are looking for students who are not just academically successful but who also work well with others and care about improving and serving surrounding communities.
9. Which concluding sentence should replace sentence 22 to provide the **best** support for the argument presented in the passage?
- A. In every city, there are homeless shelters, food pantries, youth centers, and political campaigns that depend on the support and commitment of hardworking young people.
 - B. Clearly, students who volunteer will experience a host of benefits, such as learning new skills, meeting interesting people, opening up opportunities for the future, and gaining a sense of satisfaction.
 - C. When students carefully evaluate their activities and prioritize volunteerism, they find a way to work it into their schedule and are soon encouraging their peers to do the same.
 - D. Whether they are seeking admission to college, preparing to enter the workforce, or simply looking to reduce stress, high school students who volunteer can help themselves as much as they help others.

CONTINUE TO THE NEXT PAGE ►

Moving through Mountains

- (1) An age-old proverb says that necessity is the mother of invention. (2) Centuries of human ingenuity in the face of obstacles prove this to be true. (3) For many years the Swiss Alps, a mountain range spanning southern Switzerland and northern Italy, were such an obstacle. (4) Roads and railways had to navigate around the mountains or through winding tunnels inside the mountains, making the transportation of people and goods difficult and time-consuming. (5) In 2016 these burdens were eased with the completion of the Gotthard Base Tunnel.
- (6) Construction of the high-speed railway tunnel began in 1996. (7) The tunnel was created through the use of tunnel-boring machines, which are giant drills with a flat rotating head called a cutter head. (8) Each of the tunnel-boring machines used during the construction of the tunnel was about the length of four football fields arranged end-to-end. (9) During the seventeen-year construction period, 28 million tons of rock were removed, enough to rebuild the Great Pyramid of Giza five times. (10) This massive construction project is reported to have cost \$12 billion. (11) After that, 4 million cubic meters of concrete, or enough concrete to build eighty-four Empire State Buildings, were used to construct and support the tunnel.
- (12) In a few years, the high-speed railway will carry more than 250 freight trains and 55 passenger trains a day, with most traveling at speeds of around 100 to 125 miles per hour. (13) It will be faster for people to travel between northern and southern Europe. (14) The travel time between the European cities of Zurich, Switzerland, and Milan, Italy, will be reduced by an hour. (15) Many European leaders compare the Gotthard Base Tunnel to the Channel Tunnel, a 33-mile underwater tunnel that connects the United Kingdom and France. (16) While there is no roadway in the Channel Tunnel, people can drive their cars onto special trains that will carry vehicles through to the other side.
- (17) Just as traffic congestion in major cities led to the construction of underground local transportation, natural formations, such as mountain ranges, have also sent people underground for faster, easier, and cheaper methods of transportation across larger areas. (18) There is renewed interest in constructing innovative methods of transportation that will help eliminate problems associated with traveling to and from certain areas.

- 10.** Which sentence should be added after sentence 5 to introduce the main topic of the passage?
- E.** The construction of the Gotthard Base Tunnel was approved by Swiss voters in 1992 and was funded by tolls, fuel taxes, and government loans.
 - F.** Leaders from several European countries attended the opening ceremonies for the Gotthard Base Tunnel, a Swiss tunnel.
 - G.** The Gotthard Base Tunnel is the world’s longest and deepest railway tunnel, stretching 35.5 miles straight through the base of the Swiss Alps.
 - H.** The Gotthard Base Tunnel continues to help reduce the number of freight trucks on the roadways in the Swiss Alps.
- 11.** Which sentence should be added to follow and support sentence 7?
- A.** The tunnel-boring machine is helpful to tunnel builders in the modern era and has been an improvement over dynamite.
 - B.** These enormous tunnel-boring machines function somewhat like a cheese grater, with the cutter head grinding slowly through rock and stone.
 - C.** Engineers had considered making a tunnel under the mountains for many years, but it was impossible to do without modern tunnel-boring machines.
 - D.** Different types of cutter heads are used with tunnel-boring machines depending on the geology of the area where the tunnel is being created.
- 12.** Where should sentence 11 be moved in order to improve the organization of the second paragraph (sentences 6–11)?
- E.** to the beginning of the paragraph (before sentence 6)
 - F.** between sentences 6 and 7
 - G.** between sentences 8 and 9
 - H.** between sentences 9 and 10
- 13.** Which sentence presents information that shifts away from the main topic of the third paragraph (sentences 12–16) and should be removed?
- A.** sentence 13
 - B.** sentence 14
 - C.** sentence 15
 - D.** sentence 16

- 14.** Which transition phrase should be added to the beginning of sentence 18?
- E.** Although the Gotthard Base Tunnel is mainly for freight trains,
 - F.** With the Gotthard Base Tunnel taking ten years to complete,
 - G.** Because of the successful completion of the Gotthard Base Tunnel,
 - H.** As the number of trains using the Gotthard Base Tunnel increases,
- 15.** Which concluding sentence should be added after sentence 18 to support the topic presented in the passage?
- A.** There is proof that underground tunnels such as the Gotthard Base Tunnel are beneficial to the economy of the surrounding area.
 - B.** The Gotthard Base Tunnel is an extraordinary example of how human ingenuity and persistence can overcome great obstacles.
 - C.** The completion of the Gotthard Base Tunnel shows that people can work together to achieve important goals.
 - D.** The Swiss government is confident that the economic impact of the Gotthard Base Tunnel will be worth its construction cost.

READING COMPREHENSION

QUESTIONS 16–57

DIRECTIONS: Read each of the following five texts and answer the related questions. You may write in your test booklet as needed to take notes. You should reread relevant parts of each text, while being mindful of time, before marking the **best** answer for each question. Base your answers only on the content within the text.

CONTINUE TO THE NEXT PAGE ►

Excerpt from “How Exercise Could Help You Learn a New Language”

by Gretchen Reynolds

- 1 Many scientists suspect that exercise alters the biology of the brain in ways that make it more malleable and receptive to new information, a process that scientists refer to as plasticity.
- 2 But many questions have remained unanswered about movement and learning, including whether exercise is most beneficial before, during or after instruction and how much and what types of exercise might be best.
- 3 So for a new study, which was published recently in *PLOS One*, researchers in China and Italy decided to home in on language learning and the adult brain.
- 4 Language learning is interesting. As young children, almost all of us picked up our first language easily. We didn’t have to be formally taught; we simply absorbed words and concepts.
- 5 But by early adulthood, the brain generally begins to lose some of its innate language capability. It displays less plasticity in areas of the brain related to language. As a result, for most of us, it becomes harder to learn a second language after childhood.
- 6 To see what effects exercise might have on this process, the researchers first recruited 40 college-age Chinese men and women who were trying to learn English. The students had some facility with this second language but were far from proficient.
- 7 The researchers then divided the students into two groups. Those in one group would continue to learn English as they had before, primarily while seated in rote vocabulary-memorization sessions.
- 8 The others would supplement these sessions with exercise.
- 9 Specifically, the students would ride exercise bikes at a gentle pace (about 60 percent of their maximum aerobic capacity) beginning 20 minutes before the start of the lessons and continuing throughout the 15 minutes or so of instruction.
- 10 Both groups learned their new vocabulary by watching words projected onto large screens, together with comparable pictures, such as “apple” and a Red Delicious. They were shown 40 words per session, with the sequence repeated several times.
- 11 Afterward, the students all rested briefly and then completed a vocabulary quiz, using computer keys to note as quickly as possible whether a word was with its correct picture. They also responded to sentences using the new words, marking whether the sentences were accurate or, in the case of “The apple is a dentist,” nonsensical. Most linguists¹ feel that understanding sentences shows greater mastery of a new language than does simple vocabulary improvement.
- 12 The students completed eight vocabulary sessions over the course of two months.

¹**linguists:** scientists who study language

- 13 And at the end of each lesson, the students who had ridden bikes performed better on the subsequent vocabulary tests than did the students who sat still.
- 14 They also became more proficient at recognizing proper sentences than the sedentary students, although that difference did not emerge until after several weeks of instruction.
- 15 Perhaps most interesting, the gains in vocabulary and comprehension lingered longest for the cyclists. When the researchers asked the students to return to the lab for a final round of testing a month after the lessons—without practicing in the meantime—the cyclists remembered words and understood them in sentences more accurately than did the students who had not moved.
- 16 “The results suggest that physical activity during learning improves that learning,” says Simone Sulpizio, a professor of psychology and linguistics at the University Vita-Salute San Raffaele in Milan, Italy, and a study co-author.
- 17 These improvements extend beyond simply aiding in memorization, she added. The exercise also deepened language learners’ grasp of how to use their newly acquired words.
- 18 This study involved college students performing relatively light exercise, though, and cannot tell us whether other people completing other types of exercise would achieve the same results.
- 19 It also offers no clues about what is occurring inside the brain that might be contributing to the benefits of the exercise. But many past studies have shown that exercise prompts the release of multiple neurochemicals in the brain that increase the number of new brain cells and the connections between neurons, Dr. Sulpizio says. These effects improve the brain’s plasticity and augment the ability to learn.
- 20 From a real-world standpoint, the study’s implications might seem at first to be impractical. Few classrooms are equipped with stationary bicycles. But specialized equipment is probably unnecessary, Dr. Sulpizio says.
- 21 “We are not suggesting that schools or teachers buy lots of bicycles,” she says. “A simpler take-home message may be that instruction should be flanked by physical activity.”

From “How Exercise Could Help You Learn a New Language” by Gretchen Reynolds from THE NEW YORK TIMES, August 16, 2017. Copyright © 2017 The New York Times Company.

16. How does the author’s use of comparison in paragraphs 4 and 5 contribute to the development of ideas in the excerpt?

- E. It identifies why many adults are never able to learn a second language.
- F. It demonstrates why the study of adult language learning tries to include physical activity.
- G. It highlights why more is known about language learning in children than about language learning in adults.
- H. It explains why adults were the subjects of the study rather than children.

17. Read this sentence from paragraph 6.

To see what effects exercise might have on this process, the researchers first recruited 40 college-age Chinese men and women who were trying to learn English.

Which statement describes the function of the sentence in the overall structure of the excerpt?

- A. It indicates a shift from describing the results of the study to describing the process.
- B. It introduces the aspect of the research that was most critical to determining the results.
- C. It marks the change from explaining why the study was conducted to explaining how the study was conducted.
- D. It shows how questions were raised that could be addressed in further research.

18. Which sentence is the **best** summary of the research procedure used in the study in the excerpt?

- E. Researchers had a group of non-English-speaking students learn new English vocabulary by matching words and images on a computer screen, and also by analyzing words in context; some students exercised and some students sat still while learning.
- F. Researchers first divided non-English-speaking students into two groups, with one group exercising before and while studying English and the other group studying English without exercising; then, after instruction, student learning was assessed.
- G. Researchers first had non-English-speaking students sit on stationary bikes and pedal for 20 minutes before instruction and 15 minutes during instruction; then, after a rest period, students indicated whether English sentences made sense.
- H. Researchers asked non-English-speaking students to study English while sitting or biking; then the students’ vocabulary skills were assessed after a short break period and also after a break of a full month without their having studied any of the materials.

19. In the study described in the excerpt, researchers asked English learners to assess the sense of words in context because that task is
- A. better at determining the capability learners have of language than simple vocabulary recall.
 - B. consistent with the way earlier research was conducted on adult language learners.
 - C. a better way for learners to absorb the meaning of individual vocabulary words.
 - D. more difficult for learners to perform after an extended period of time has passed.
20. Read these sentences from paragraph 17.

These improvements extend beyond simply aiding in memorization, she added. The exercise also deepened language learners' grasp of how to use their newly acquired words.

Which sentence from the excerpt **best** explains why Sulpizio feels confident in the conclusion stated in these sentences?

- E. "They were shown 40 words per session, with the sequence repeated several times." (paragraph 10)
- F. "They also responded to sentences using the new words, marking whether the sentences were accurate or, in the case of 'The apple is a dentist,' nonsensical." (paragraph 11)
- G. "And at the end of each lesson, the students who had ridden bikes performed better on the subsequent vocabulary tests than did the students who sat still." (paragraph 13)
- H. "They also became more proficient at recognizing proper sentences than the sedentary students, although that difference did not emerge until after several weeks of instruction." (paragraph 14)

21. Paragraph 18 shows that the author’s perspective toward the study described in the excerpt is that
- A. though the study allowed some conclusions to be drawn, these conclusions are limited.
 - B. while the study focused on moderate exercise, it would have been better to focus on difficult exercise.
 - C. while the study was performed appropriately, it is doubtful the results can be duplicated.
 - D. though the study was about learning language, its conclusions can be applied to learning in general.
22. Which evidence from the excerpt is **most** relevant to the author’s claim that there are many unanswered questions about the relationship between movement and learning?
- E. the method of the study using one type of exercise and people from one age group
 - F. the statement from Sulpizio explaining past studies on neurochemicals in the brain
 - G. the finding of the study that exercise helps people recall information over a long period of time
 - H. the statement from Sulpizio describing how the results of the study can be applied

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This narrative is about the Nez Perce, an American Indian tribe, in what is now northern Idaho. The tribe is preparing for a gathering before the coming winter.

Excerpt from *Do Them No Harm!*

by Zoa L. Swayne

- 1 In the moon of Ta-Yum, the hottest days of summer, when salmon spawn in the little streams and huckleberries ripen in the high mountains, people from many villages of the Chopunnish Nation gathered in the Oyaip Prairie for the work and festivities of their annual camas¹ harvest.
- 2 From far and near The People came. . . .
- 3 Should a stranger enter their homeland and ask, "Where are you from?" the reply was always, "We are Nee-mee-poo, The People who live here in this place."
- 4 It was a time of Lawtiwa-mah-ton—a time of being friends together—when The People came for this last chance to enjoy being together before the Cold Moons kept them close to their fires. The visiting and trading, the foot racing and horse racing, and the gambling and stick games would be remembered and talked about long after they had forgotten the drudgery of digging and roasting camas, picking berries, or drying meat and fish. Lawtiwa-mah-ton! It was good to be friends together.
- 5 As was their custom since wahk-kee-ma, a time far back beyond the memory of man, they set up their camps in the same locations their parents and grandparents had occupied before them. Red Bear's people, from Kamiah, made their camp near the trail that came out of the mountains. Their neighbors in Kamiah Valley, The People from Tee-e-lap-a-lo, had their camp close by. Across the wide meadow, by the great roasting pits, the camps of the Te-wap-poo and Ask-kah-poo were located. The tepees and ish-nash, brush shelters, of other groups nestled in their accustomed areas in and among the pines in such numbers that they encircled the entire meadow land.
- 6 Red Bear's people had traveled all summer with neighboring bands, gathering and preparing roots, picking and drying berries, drying and smoking meat and fish for their winter food supply. Now they were at the Oyaip camp. The women worked hard to dig and cure as many bags of roots as they could during the warm, sunny days, for the sharp night air brought warnings that WARM was going and COLD was coming.
- 7 Everyone helped in some way. Most of the men fished or hunted for meat. While many of the women dug and roasted camas, other women and older children picked and dried berries.
- 8 And the younger children played. They played at hunting. They played with the babies. They played with their horses and puppies. They learned how to live through their play.
- 9 This sun, happiness, peace, and quiet blessed the Red Bear camp. All were busy with their daily tasks, until sudden cries came from the children playing by the trail.

¹**camas:** a wild, edible plant

- 10 "People coming! People coming! People coming on the trail from the high mountains!" they called as they ran to their elders, who looked sharply at the figures of approaching horsemen.
- 11 Were they friends or enemies? Did they bring good news or bad?
- 12 "Who can it be? What brings them here?" were the questions in every mind.
- 13 "Could they be the four hunters who had gone to Buffalo Country two summers past? Would they have news of the families who had gone long ago to Buffalo Country and never returned?"
- 14 It was customary for a hunting party to be gone for more than one season.
- 15 "Looks like hunters," the older men agreed. "Looks like they had good hunting. Maybe our four hunters. [They have] been gone many moons."
- 16 "Looks like five people—not four," others observed.
- 17 Excitement grew as the riders came close enough to be recognized.
- 18 "A-a-a-a-a, they are our four hunters! But who is the fifth person?" they asked.
- 19 "Looks like a woman. Who is she?"
- 20 The hunters rode up to the welcoming crowd, proud to show off the loads of meat, hides, and other trophies of their hunt. They paraded around the encampment for all to see how strong their Hunting Power had been—what great hunters they, themselves, were.
- 21 Red Bear's people rejoiced at their hunters' success. Good hunters brought good to everybody. The meat meant plenty of food and the hides meant soft-tanned robes to give comfort through the Cold Moons. But it was the sight of the frail figure of the woman that aroused their curiosity. Who was she? Where had she come from? . . .
- 22 "Belongs to Red Bear people. Gone then come back," the hunters said, as they dismounted and unloaded their packs. . . .
- 23 Now they could see! She was the daughter of the family gone so long ago! The girl-child who had left came back now—a grown woman.
- 24 "Wat-ku-ese!" the women cried. "Gone-from-Home-then-Come-Back. Wat-ku-ese!" And Wat-ku-ese was her name from that time on.
- 25 Gentle arms lifted Wat-ku-ese from her horse. The women brought her food and made a place for her to rest. For many suns they cared for her until she became stronger.
- 26 One evening Wat-ku-ese told her story for all to hear.

From DO THEM NO HARM!: Lewis and Clark Among the Nez Perce by Zoa L. Swayne. Published by Caxton Press. Copyright © 1990 by Zoa L. Swayne. Orofino, Idaho and Legacy House, Inc. Orofino, Idaho. All rights reserved.

23. Paragraph 1 contributes to the setting of the excerpt by establishing that
- A. the story's location is a plentiful place that allows The People to enjoy a comfortable gathering.
 - B. the events in the story occurred in the past and are being remembered by The People.
 - C. the camp in the story is changing and that the changes are causing problems for The People.
 - D. the story's plot begins as The People are observing their land and what it offers them.
24. In paragraphs 4 and 6, the beginning of the change in seasons affects the characters **mainly** by
- E. causing them to collect their bounty and feel eager to celebrate before it becomes cold.
 - F. making them want to rest and relax before the challenging work of the harvest begins.
 - G. forcing them to give up leisure time to prepare their camp for the coming winter.
 - H. allowing them to trade the goods they have prepared during the warm months.
25. How do paragraphs 7 and 9 convey a central idea of the excerpt?
- A. by explaining that The People often explore different areas, which shows the value of trying new things
 - B. by revealing that The People must complete many tasks, which shows the necessity of being organized
 - C. by suggesting that The People are influenced by the weather, which shows their close relationship with nature
 - D. by demonstrating that The People work together, which shows the importance of contributing to the community
26. In paragraph 8, the author repeats the word "played" **most likely** to
- E. emphasize how much the adults enjoy observing the children.
 - F. show how everyone finds a way to enjoy being at the gathering.
 - G. indicate the variety of activities available at the gathering.
 - H. characterize the manner in which the children master the work of adults.

27. How do paragraphs 11–13 affect the plot of the excerpt?
- A. They establish a feeling of mystery by presenting different points of view about the arriving people.
 - B. They contribute to the rising action by developing the idea that the arriving people could present a problem.
 - C. They lead to a turning point by describing the moment that the crowd realizes why the strangers have come.
 - D. They explain the cause of the main conflict by providing background information about the strangers.
28. In paragraph 20, the phrases “trophies of their hunt” and “paraded around” affect the paragraph by
- E. suggesting that the hunters are rewarded for their leadership.
 - F. emphasizing that The People are dependent on the hunters for food.
 - G. revealing that the hunters are pleased with their success.
 - H. indicating that The People have gathered specifically to welcome the hunters.
29. Which quotation from the excerpt **best** supports the idea that The People feel a connection to members of their group in spite of distance?
- A. “As was their custom since wahk-kee-ma, a time far back beyond the memory of man, they set up their camps in the same locations their parents and grandparents had occupied before them.” (paragraph 5)
 - B. “ ‘Would they have news of the families who had gone long ago to Buffalo Country and never returned?’ ” (paragraph 13)
 - C. “ ‘Looks like hunters,’ the older men agreed. ‘Looks like they had good hunting.’ ” (paragraph 15)
 - D. “But it was the sight of the frail figure of the woman that aroused their curiosity.” (paragraph 21)

30. Read these sentences from paragraph 22.

“Belongs to Red Bear people. Gone then come back.”

How does this statement affect the villagers in the excerpt?

- E.** It increases their concern about why the woman is in their camp.
- F.** It makes them hopeful that more lost members will return before the winter.
- G.** It causes shock, leading them to grieve for the loss of the woman and her family.
- H.** It inspires amazement, making them want to reconnect with their returned family member.

31. The details in paragraph 5 about the locations of the camps and the details in paragraphs 23–24 about the reaction to the woman convey a theme of the excerpt by

- A.** revealing the difficulties posed by moving often each year.
- B.** showing the connectedness among The People through the years.
- C.** indicating the value to The People of retelling stories about the past.
- D.** demonstrating the importance of choosing appropriate names.

CONTINUE TO THE NEXT PAGE ►

Samuel Morse, an American inventor, is credited with creating the electronic telegraph, a communication device that allows users to send messages using a system of short and long pulses that represent letters, numbers, and punctuation. In 1844, the United States Congress passed the Telegraph Bill, which provided Morse with the funds to build an electric telegraph system.

Invention of the Telegraph

Earlier Signal Systems

- 1 Long before Samuel F. B. Morse electrically transmitted his famous message “What hath God wrought?” from Washington to Baltimore on May 24, 1844, there were signaling systems that enabled people to communicate over distances. Most were visual or “semaphore” systems using flags or lights. In the eighteenth century, such systems used an observer who would decipher a signal from a high tower on a distant hill and then send it on to the next station. The young American republic wanted just such a system along its entire Atlantic coast and offered a prize of \$30,000 for a workable proposal. The framers of this legislation¹ had no way of knowing that when they used the word “telegraph” to refer to this visual semaphore system, they would be offered an entirely new and revolutionary means of communication—electricity.

The Growth of an Idea

- 2 The idea of using electricity to communicate over distance is said to have occurred to Morse during a conversation aboard ship when he was returning from Europe in 1832. Michael Faraday’s recently invented electromagnet was much discussed by the ship’s passengers, and when Morse came to understand how it worked, he speculated that it might be possible to send a coded message over a wire. While a student at Yale College years before, he had written his parents a letter about how interesting he found the lectures on electricity. Despite what he had learned at Yale, Morse found when he began to develop his idea that he had little real understanding of the nature of electricity, and after sporadic attempts to work with batteries, magnets, and wires, he finally turned for help to a colleague at the University of the City of New York, Leonard D. Gale.
- 3 Gale was a professor of chemistry and familiar with the electrical work of Princeton’s Joseph Henry, a true pioneer in the new field. Well before Morse had his shipboard idea about a telegraph, Henry rang a bell at a distance by opening and closing an electric circuit. In 1831, he had published an article, of which Morse was unaware, that contained details suggesting the idea of an electric telegraph. Gale’s help and his knowledge of this article proved crucial to Morse’s telegraph system because Gale not only pointed out flaws in the system but showed Morse how he could regularly boost the strength of a signal and overcome the distance problems he had encountered by using a relay system Henry had invented. Henry’s experiments, Gale’s assistance, and, soon after, hiring the young technician Alfred Vail were keys to Morse’s success.

Obstacles and Opportunities

- 4 By December 1837, Morse had enough confidence in his new system to apply for the federal government’s appropriation, and during the next year he conducted demonstrations of his telegraph both in New York and Washington.

¹**legislation:** Telegraph Bill

- 5 However, when the economic disaster known as the Panic of 1837 took hold of the nation and caused a long depression, Morse was forced to wait for better times. It was during this period that Morse visited Europe again and tried not only to secure patent protection overseas but to examine competing telegraph systems in England. . . .
- 6 By 1843, the country was beginning to recover economically, and Morse again asked Congress for the \$30,000 that would allow him to build a telegraph line from Washington to Baltimore, forty miles away. The House of Representatives eventually passed the bill containing the Morse appropriation, and the Senate approved it in the final hours of that Congress's last session. With President Tyler's signature, Morse received the cash he needed and began to carry out plans for an underground telegraph line.

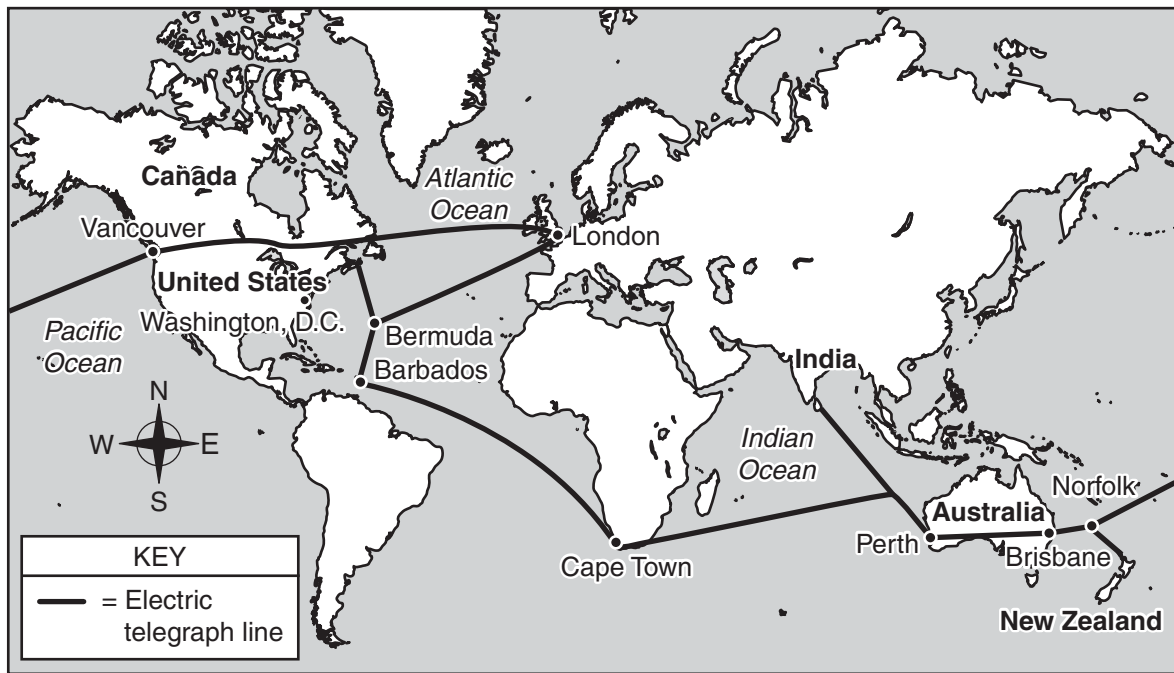
Realizing a Great Invention

- 7 Morse had hired the ingenious construction engineer Ezra Cornell to lay the pipe carrying the wire, and although Cornell did his job superbly, one of Morse's partners, Congressman F. O. J. Smith, had purchased wire with defective insulation. Too much time had been wasted laying bad wire, and with the project on a rigid deadline, something had to be done quickly. Cornell suggested that the fastest and cheapest way of connecting Washington and Baltimore was to string wires overhead on trees and poles. The desperate Morse gave the go-ahead, and the line was completed in time for the dramatic and spectacularly successful link between the Supreme Court chamber of the Capitol building and the railroad station in Baltimore.
- 8 Soon, as overhead wires connected cities up and down the Atlantic coast, the dots-and-dashes method² that recorded messages on a long moving strip of paper was replaced by the operator's ability to interpret the code in real time. . . . Telegraph lines soon extended westward, and within Morse's own lifetime they connected the continents of Europe and America.

"Invention of the Telegraph"—Public Domain/Library of Congress

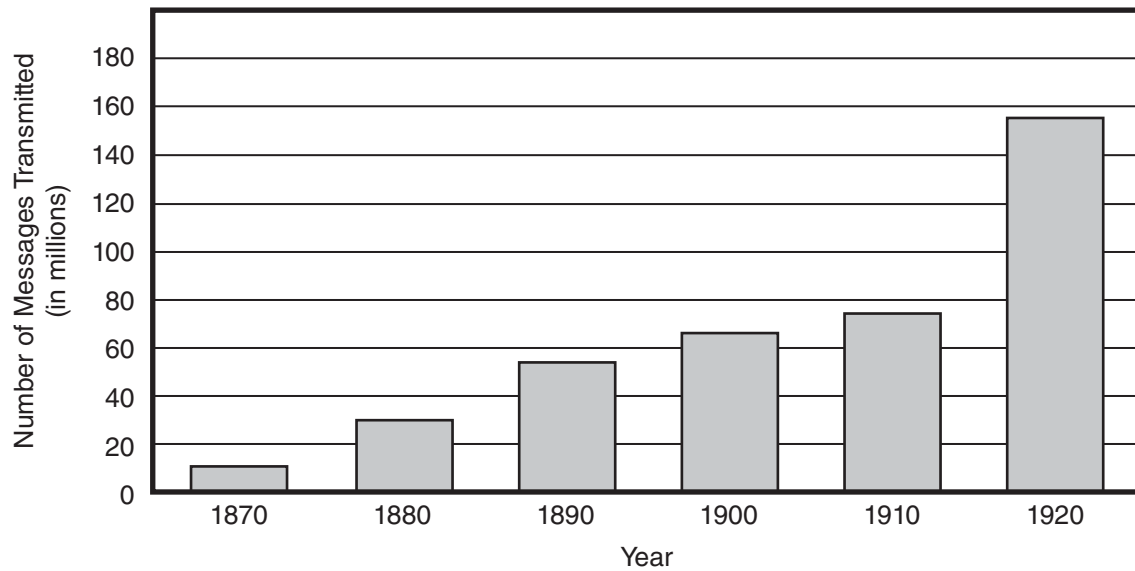
²**dots-and-dashes method:** the short and long pulses of Morse code that are sent and received by telegraph operators

BRITISH ALL-RED TELEGRAPH LINE, 1902



The first transatlantic electric telegraph message was sent in 1858, and by 1902 the British All-Red Line connected most of the world.

TELEGRAPH MESSAGES TRANSMITTED, 1870–1920



Source: U.S. Bureau of the Census. *Historical Statistics of the United States, Colonial Times to 1970, Bicentennial Edition, Part 2*. Washington, D.C., 1975.

- 32.** Which statement describes how the author’s use of problem-solution in paragraph 2 contributes to the development of ideas in the passage?
- E.** Morse’s discussion on a ship about Faraday’s electromagnet reminded him of the Yale College lectures on electricity, which he had enjoyed but had not fully understood, inspiring him to learn more about electricity from his colleague Gale.
 - F.** Morse’s discouragement over his lack of knowledge of electricity prompted him to experiment with batteries, magnets, and wires, which led to the development of a new long-distance communication system.
 - G.** Morse’s difficulty in understanding how Faraday’s electromagnet worked was frustrating, and it pushed him to create a system for sending signals over wires.
 - H.** Morse’s longtime fascination was not enough to make up for his lack of knowledge about electricity, so he eventually sought help from Gale.
- 33.** The details of the section “The Growth of an Idea” convey a central idea of the passage by suggesting that
- A.** the collaborative efforts of colleagues resulted in successful communication over a wire.
 - B.** a great deal of interest and work was devoted to understanding how to use electricity to send signals.
 - C.** Faraday’s invention of the electromagnet inspired the invention of the telegraph.
 - D.** colleges like Yale played a great role in making new discoveries about electricity and its applications.
- 34.** Which statement describes how the author’s use of sequencing in paragraph 3 contributes to the overall structure of the passage?
- E.** It shows that several people were simultaneously attempting to create an electric telegraph.
 - F.** It shows how the invention of the electric telegraph depended on information and techniques discovered by others.
 - G.** It shows that multiple means of long-distance communication were being used at the same time.
 - H.** It shows how quickly long-distance communication changed from visual signals to electrical signals.

35. Which sentence is the **best** summary of how Morse obtained the funding necessary to build his telegraph system?

- A. Morse’s application for a federal grant was delayed until 1843, so he spent time traveling in Europe, where he concentrated on obtaining a patent for his system.
- B. Morse applied for a government grant that required both houses of Congress and the president to pass a bill awarding him \$30,000 for his telegraph project.
- C. Morse applied for a government appropriation and conducted telegraph demonstrations to show that his system could work, and after a delay caused by a financial depression, Congress approved the \$30,000 appropriation in 1843.
- D. Working with Gale and Vail allowed Morse to find flaws in Henry’s work and to develop his own ideas before applying for the federal government appropriation.

36. Read this sentence from paragraph 7.

The desperate Morse gave the go-ahead, and the line was completed in time for the dramatic and spectacularly successful link between the Supreme Court chamber of the Capitol building and the railroad station in Baltimore.

The words “dramatic” and “spectacularly” in the sentence convey a

- E. sense of relief and fulfillment that the line was finished.
- F. sense of wonder and celebration that the telegraph line was accomplished.
- G. feeling of excitement about the future possibilities of the telegraph.
- H. feeling of confidence about being able to continue the work.

37. How does the graph support the ideas in paragraph 8?

- A. It indicates how welcome the improvement of long-distance communication was in the United States.
- B. It provides evidence of the dramatic increase in the number of telegraph messages as Morse’s system expanded across the United States.
- C. It reveals that by the twentieth century millions of people had used the telegraph despite earlier hesitations about the system.
- D. It shows how improvements that allowed Morse code to be read in real time made relaying telegraph messages faster and increased the system’s usage.

38. The idea that, in the mid-nineteenth century, the United States was mostly unaware of the possibilities of electricity is illustrated in the passage **mainly** through the
- E. description of the government's initial desire to expand a semaphore signaling system that used either flags or lights along the Atlantic Coast.
 - F. delay by the House of Representatives to pass the bill funding Morse's telegraph line six years after he first applied for the appropriation.
 - G. discussions of the newly invented electromagnet that sparked the idea of sending codes through wires.
 - H. description of how an electric circuit could be closed to ring a bell at a distance.
39. With which statement would the author **most likely** agree?
- A. Morse's telegraph was successful because the wires were strung aboveground rather than underground as originally planned.
 - B. Understanding the importance of Morse's telegraph requires detailed knowledge of electrical systems.
 - C. The implementation of Morse's telegraph system was overly influenced by economic factors.
 - D. Morse's invention of the telegraph made a great stride toward better connecting people across the United States and across the world.
40. How does the map provide additional support for a central idea of the passage?
- E. by demonstrating that Morse's telegraph system greatly exceeded the limitations of previous long-distance communication systems
 - F. by indicating that Morse was lacking foresight by seeking a patent only in North America
 - G. by proving that Gale's advice helped Morse extend the telegraph's range much farther than first thought possible
 - H. by showing that the Morse telegraph had a surprisingly immense influence across the world

Snowy Mountains

by John Gould Fletcher

- Higher and still more high,
Palaces made for cloud,
Above the dingy city-roofs
Blue-white like angels with broad wings,
5 Pillars of the sky at rest
The mountains from the great plateau
Uprise.
- But the world heeds them not;
They have been here now for too long a time.
10 The world makes war on them,
Tunnels their granite cliffs,
Splits down their shining sides,
Plasters their cliffs with soap-advertisements,
Destroys the lonely fragments of their peace.
- 15 Vaster and still more vast,
Peak after peak, pile after pile,
Wilderness still untamed,
To which the future is as was the past,
Barrier spread by Gods,
20 Sunning their shining foreheads,
Barrier broken down by those who do not need
The joy of time-resisting storm-worn stone,
The mountains swing along
The south horizon of the sky;
25 Welcoming with wide floors of blue-green ice
The mists that dance and drive before the sun.

“Snowy Mountains” by John Gould Fletcher—Public Domain

41. The description in the first stanza (lines 1–7) helps establish a central idea of the poem by
- A. comparing the length of time the mountains have existed with the length of time the city has existed.
 - B. contrasting the grandeur of the mountains with the structures in the city below them.
 - C. implying that the mountains are a source of inspiration to the people in the city below.
 - D. suggesting that the mountains are larger than the people in the city realize.

42. Read line 5 from the poem.

Pillars of the sky at rest

The line helps develop the theme of the poem by suggesting that the mountains

- E. serve a noble and supportive purpose in the world.
 - F. attract the clouds with their strength and permanence.
 - G. remain untamed through the ages.
 - H. provide protection for the people.
43. How does isolating the word “Uprise” in line 7 affect the meaning of the poem?
- A. It creates a contrast between the great plateau and the city buildings.
 - B. It reveals the similarity between the tall buildings in the city and the tall mountains on the horizon.
 - C. It creates a vision of the region before people developed the land.
 - D. It emphasizes that the mountains dominate the landscape.
44. Which line from the poem **best** supports the idea that people have sacrificed priceless natural beauty in order to make a profit?
- E. “Above the dingy city-roofs” (line 3)
 - F. “The world makes war on them,” (line 10)
 - G. “Tunnels their granite cliffs,” (line 11)
 - H. “Plasters their cliffs with soap-advertisements,” (line 13)

45. How does the poet develop the speaker's point of view in the second stanza (lines 8–14)?
- A. by describing images of the mountains' awe-inspiring size and strength
 - B. by illustrating the differences among the various ways humans can affect the natural environment
 - C. by criticizing society for taking careless, harmful courses of action against nature
 - D. by demonstrating how the mountains and the people are able to benefit from each other
46. Which detail from the poem reflects the speaker's view that people often fail to appreciate what is familiar?
- E. "The mountains from the great plateau" (line 6)
 - F. "They have been here now for too long a time." (line 9)
 - G. "Splits down their shining sides," (line 12)
 - H. "To which the future is as was the past," (line 18)
47. Read lines 21–22 from the poem.

**Barrier broken down by those who do not need
The joy of time-resisting storm-worn stone,**

How do the lines help convey the speaker's point of view?

- A. They suggest that the speaker wants to remove the obstacles that prevent others from experiencing the wonders of nature.
- B. They reveal the speaker's opinion that some people are too busy to appreciate natural beauty.
- C. They reflect the speaker's dismay that people destroy the natural landscape without understanding the ramifications of their actions.
- D. They explain that the speaker is confident that nature will never be fully destroyed by people.

48. How do the details in the third stanza (lines 15–26) **most** contribute to the development of a theme of the poem?

- E.** by reflecting nature’s capacity to resist change
- F.** by showing that nature is capable of influencing human will
- G.** by exposing how a lack of awareness leads to nature’s ruin
- H.** by explaining why people must respect nature

49. Read lines 23–26 from the poem.

**The mountains swing along
The south horizon of the sky;
Welcoming with wide floors of blue-green ice
The mists that dance and drive before the sun.**

The personification in these concluding lines of the poem suggests that the mountains are

- A.** gracious hosts who are untroubled by the actions of people.
- B.** unaware of their coming destruction.
- C.** lively entertainers who are amused by the everyday concerns of people.
- D.** too proud to reveal their pain.

Excerpt from “The Past and the Future of the Earth’s Oldest Trees”

by Alex Ross

- 1 About forty-five hundred years ago, not long after the completion of the Great Pyramid at Giza, a seed of *Pinus longaeva*, the Great Basin bristlecone pine, landed on a steep slope in what are now known as the White Mountains, in eastern California. The seed may have travelled there on a gust of wind, its flight aided by a winglike attachment to the nut. Or it could have been planted by a bird known as the Clark’s nutcracker, which likes to hide pine seeds in caches; nutcrackers have phenomenal spatial¹ memory and can recall thousands of such caches. This seed, however, lay undisturbed. On a moist day in fall, or in the wake of melting snows in spring, a seedling appeared above ground—a stubby one-inch stem with a tuft of bright-green shoots.
- 2 Most seedlings die within a year; the mortality rate is more than ninety-nine percent. The survivors are sometimes seen growing in the shadow of a fallen tree. The landscape of the Ancient Bristlecone Pine Forest, as this area of the White Mountains is called, is littered with fragments of dead trees—trunks, limbs, roots, and smaller chunks. *Pinus longaeva* grows exclusively in subalpine regions of the Great Basin, which stretches from the eastern slopes of the Sierra Nevada to the Wasatch Range, in Utah. Conditions are generally too arid for the dead wood to rot; instead, it erodes, sanded down like rock. The remnants may harbor nutrients and fungi that help new trees grow. Bristlecones rise from the bones of their ancestors—a city within a cemetery.
- 3 Coast redwoods and giant sequoias, California’s gargantuan world-record-holding trees, can grow fifty feet or more in their first twenty years. Bristlecones rise agonizingly slowly. After four or five years, the seedling on the steep slope would have been just a few inches higher, sprouting needles in place of the embryonic shoots. The needles are a deep green, tough, resinous, and closely bunched in groups of five. On a mature tree, they live for fifty years or more. Decades may have passed before the tree was human height, and decades more before it resembled a conventional pine. Bristlecone saplings grow straight up, with relatively sparse foliage, looking like undernourished Christmas trees. After a few hundred years—by which time the Old Kingdom of Egypt had fallen—it was probably forty or fifty feet in height.
- 4 Many tree species live for hundreds of years. A smaller but not inconsiderable number, including the sequoias and certain yews, oaks, cypresses, and junipers, survive for thousands. Once a bristlecone has established itself in the unforgiving conditions of the White Mountains, it can last almost indefinitely. The trees tend to grow some distance from one another, so fires almost never destroy an entire stand. Because only a few other plant species can handle the dry, cold climate, the bristlecones face little competition. Unlike most plants, they tolerate dolomite soil, which is composed of a chalky type of limestone that is heavily alkaline and low in nutrients. As for insect threats, bristlecone wood is so dense that mountain-pine beetles and other pests can rarely burrow their way into it.
- 5 Empires rose and fell; wars raged; . . . and the tree from 2500 B.C. continued its implacable slow-motion existence, adding about two-hundredths of an inch to the diameter of its trunk each

¹**spatial:** relating to physical space

year. Minute changes in the tree-ring record make bristlecones an exceptionally useful source of data about changing conditions on Earth. When rains are heavier than normal, the rings widen. When volcanic eruptions cause global cooling, frost rings make the anomaly² visible. . . .

- 6 As the millennia go by, bristlecones become contorted and wraithlike.³ The main stem, or leader, dies back. Entire branches, even the trunk itself, become fossils. At first glance, the tree may look dead. Such is the case of the forty-five-hundred-year-old tree that clings to life near the tourist path that now runs through the Ancient Bristlecone Pine Forest. Spears of dead wood jut into the air. The trunk is a marbled hulk stripped of bark, like driftwood thrown from a vanished ocean. A ribbon of live bark runs up one side, funneling water and nutrients to clumps of green needles high above. All told, the tree is an unprepossessing specimen; most people march past it without giving it a second glance. . . .
- 7 . . . No two super-elderly trees look alike, to the point where they have acquired the characteristics of individuals. Trees are prone to anthropomorphism; we project our dreams and our anxieties onto them. Bristlecones have been called elders, sentinels, sages. The possibility that climate change will cause their extinction has inspired a spate⁴ of alarmed news stories, although tree scientists tend to discount the idea that the bristlecones are in immediate danger. They have survived any number of catastrophes in the past; they may survive humanity.

From "The Past and the Future of the Earth's Oldest Trees" by Alex Ross from THE NEW YORKER, January 20, 2020. Copyright © 2020 by Condé Nast. All rights reserved.

²**anomaly:** oddity

³**wraithlike:** ghostly

⁴**spate:** sudden abundance

50. What is the effect of comparing groups of bristlecone pines to "a city within a cemetery" (paragraph 2)?
- E. It illustrates that the new tree's appearance is markedly different from that of mature bristlecone pines.
 - F. It emphasizes the new tree's ability to use resources left by prior generations of bristlecone pines.
 - G. It highlights the tree's capacity for a long life if it is able to survive to maturity.
 - H. It reveals that the tree's dense growth patterns are unusual in arid conditions.

51. In paragraph 4, the idea that the bristlecone pine can withstand difficult conditions is illustrated **mainly** through

- A. a comparison of the bristlecone pine with similar trees that are known for their long lives.
- B. a description of specific qualities of the bristlecone pine that make it suited to its environment.
- C. an explanation of why the White Mountains are the best place for the bristlecone pine to grow.
- D. an examination of why the bristlecone pine has few competitors in the White Mountains.

52. Read this sentence from paragraph 5.

Empires rose and fell; wars raged; . . . and the tree from 2500 B.C. continued its implacable slow-motion existence, adding about two-hundredths of an inch to the diameter of its trunk each year.

The sentence contributes to the development of ideas in the excerpt by

- E. emphasizing the bristlecone pine's precise growth pattern.
- F. suggesting that things that seem well established face threats.
- G. describing the pace of the life cycle of the bristlecone pine.
- H. explaining the reason for the bristlecone pine's longevity.

53. Which claim is **best** supported by the details in paragraph 5 about the bristlecone pine's tree-ring record?

- A. The bristlecone pine must be protected from extinction so that there is a record of changing Earth conditions.
- B. The structure of the bristlecone pine allows scientists to predict changing Earth conditions over time.
- C. The slow growth process of the bristlecone pine allows it to survive in spite of changing Earth conditions.
- D. The bristlecone pine can be used to gather information on changing Earth conditions because of its long life.

54. The phrases “clings to life” and “stripped of bark” in paragraph 6 are used to highlight
- E. that the tree is fragile and needs help to avoid becoming extinct.
 - F. how the tree can be misunderstood because of its appearance.
 - G. how the tree interests observers with its unusual appearance.
 - H. that the tree manages to survive even when partially fossilized.
55. The details in paragraph 7 convey a central idea of the excerpt by suggesting that
- A. the success of the bristlecone pine lies in its ability to resist climate change.
 - B. the uncommon appearance of the bristlecone pine is related to its ability to survive.
 - C. the attachment that humans have to the bristlecone pine is understandable.
 - D. the longevity of the bristlecone pine continues to encourage experts.
56. What is the **best** summary of the bristlecone pine’s life cycle?
- E. The bristlecone pine takes root as a stubby seedling in the mountains. Over time, it grows slowly in a hostile environment with alkaline soil that ensures little competition from other trees.
 - F. The bristlecone pine grows up slowly from the fragments of its dead ancestors. Over time, it begins to become contorted in appearance but continues to funnel water and nutrients to its stem.
 - G. The bristlecone pine grows slowly in an environment hostile to other trees. Over time, its trunk and some branches fossilize, but it funnels water and nutrients to other parts that are still alive.
 - H. The bristlecone pine grows only in one location in the mountains. Over time, it develops sparse foliage that fossilizes slowly, but the trunk and branches stay alive, as revealed by ribbons of live bark.
57. Throughout the excerpt, the author conveys his point of view **mainly** by
- A. sharing details about the qualities of the bristlecone pine that make it an unusual tree.
 - B. comparing the bristlecone pine to global empires that lasted thousands of years before falling.
 - C. explaining the way the seemingly sickly appearance of the bristlecone pine contrasts with its ability to survive.
 - D. arguing that the bristlecone pine has survived past threats to its survival and will continue to thrive.

PART 2 — MATHEMATICS

57 QUESTIONS

IMPORTANT NOTES

- (1) Formulas and definitions of mathematical terms and symbols are **not** provided.
- (2) Diagrams other than graphs are **not** necessarily drawn to scale. Do not assume any relationship in a diagram unless it is specifically stated or can be determined from the information given.
- (3) Assume that a diagram is in one plane unless the question specifically states that it is not.
- (4) Graphs are drawn to scale. Unless stated otherwise, you can assume relationships according to appearance. For example, lines on a graph that appear to be parallel can be assumed to be parallel. This is also true for concurrent lines, straight lines, collinear points, right angles, etc.
- (5) Reduce (simplify) all fractions to lowest terms.

GRID-IN QUESTION NOTES

- (1) For each grid-in question, write your answer at the top of the grid.
- (2) Begin recording your answer in the columns on the far left.
- (3) Fill in the circle under the box that matches the number or symbol you wrote. Leave the negative sign bubble blank if your answer is positive.

(Answer: -1.5)

Negative sign →

← Decimal point

(Answer: 3.2)

Negative sign →

← Decimal point

CONTINUE TO THE NEXT PAGE ►

GRID-IN QUESTIONS

QUESTIONS 58–62

DIRECTIONS: Solve each problem. On the answer sheet, write your answer in the boxes at the top of the grid. Start on the left side of each grid. Print only one number or symbol in each box. Under each box, fill in the circle that matches the number or symbol you wrote above.

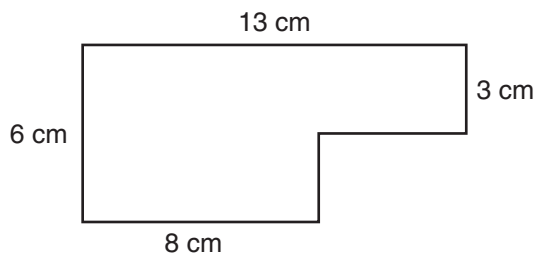
- Do not fill in a circle under an unused box.
- Do not leave a box blank in the middle of an answer.

58. $\frac{3}{5}(2x + 5) - 2x$

After the expression above is simplified, what is the coefficient of x , expressed as a decimal?

59. Two number cubes have sides labeled 1 through 6. Both number cubes are rolled. How many different outcomes will have a sum of 6?

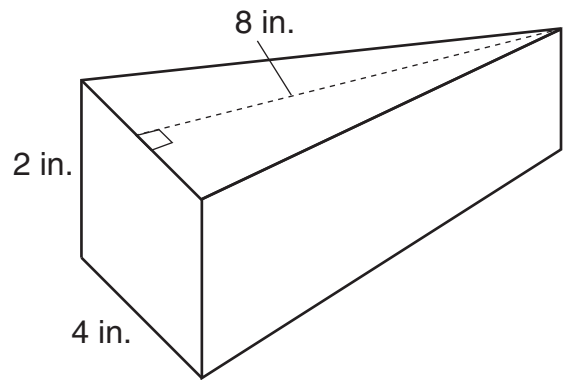
60.



The figure above shows a scale drawing of a garden, where 1 centimeter represents 2.5 meters. What is the perimeter of the actual garden in meters?

- 61.** At one location, the temperature increased from -7°F to 4°F between 5:00 a.m. and 9:00 a.m. By 11:00 a.m., the temperature was 3.5 times the 9:00 a.m. temperature. What was the total increase in temperature between 5:00 a.m. and 11:00 a.m., in degrees Fahrenheit?

- 62.** The figure below is a triangular prism. The lateral sides are rectangles.



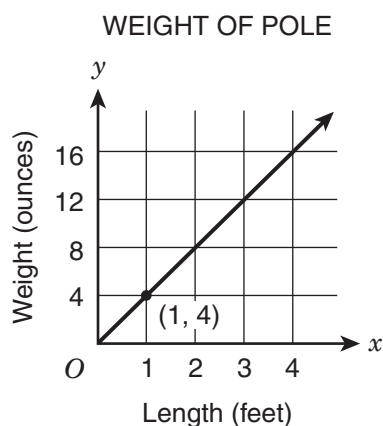
What is the volume of the figure, in cubic inches?

MULTIPLE CHOICE QUESTIONS

QUESTIONS 63–114

DIRECTIONS: Solve each problem. Select the answer from the choices given. Mark the letter of your answer on the answer sheet. When you are solving problems, you can write in the test booklet or on the scrap paper given to you.

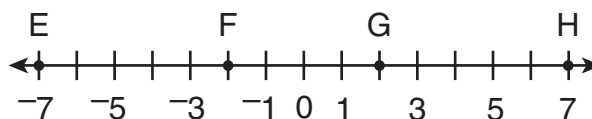
- 63.** This graph shows y , the weight of a pole of length x .



What is represented by the point with coordinates $(1, 4)$?

- A.** The unit rate is 4 ounces per foot.
- B.** The y -intercept is 4.
- C.** A pole 4 feet long weighs 1 ounce.
- D.** The length increases 4 feet for each 1 ounce of weight.

- 64.** The number line shows points E, F, G, and H.



Which point represents the sum of $2.5 + (-4.5)$?

- E.** point E
- F.** point F
- G.** point G
- H.** point H

- 65.** If $\frac{9}{2x} = \frac{3y}{8}$, where $x \neq 0$ and $y \neq 0$, what is the product of x and y ?

- A.** 4
- B.** 6
- C.** 12
- D.** 16

-
- 66.** A car dealership is having a sale where all cars have a 12% discount. What is the discounted price of a car that originally cost \$15,600?

E. \$13,728
F. \$14,300
G. \$14,400
H. \$15,588

67. $4\frac{2}{3} \div 2\frac{1}{2} =$

A. $\frac{15}{28}$
B. $1\frac{13}{15}$
C. $2\frac{1}{3}$
D. $3\frac{1}{3}$

- 68.** In a sports league, each team has 36 players and 3 coaches. There are also a number of team assistants. The ratio of team assistants to players is $\frac{1}{6}$. What is the ratio of coaches to assistants?

E. $\frac{1}{4}$
F. $\frac{1}{2}$
G. $\frac{2}{3}$
H. $\frac{5}{6}$

-
- 69.** Lena is building a fence. She will need to dig holes to help support the posts that hold up the fence. The holes need to have a depth of $3\frac{1}{3}$ feet below the ground. Each post is 10 feet long. What is the height of the part of the post that is above the ground?

- A.** $6\frac{2}{3}$ feet
B. $7\frac{1}{3}$ feet
C. 10 feet
D. $13\frac{1}{3}$ feet

70. $p + 2r = r(p + 1) + 1$

In the equation above, if $r = 2$, what is the value of p ?

- E.** 0
F. 1
G. 2
H. 3

- 71.** Karen reads at the rate of 60 pages per day. Martina reads 25% more pages per day than Karen. If the two students begin 1,500-page reading assignments on the same day, how many days sooner will Martina complete the assignment than Karen?

- A.** 4
B. 5
C. 7
D. 15

-
- 72.** Claire is running on a circular track. She has already run 650 meters. Each lap around the track is a distance of 120 meters. Her goal is to run at least 2,500 meters in all. How many more laps does she need to complete in order to reach her goal?

- E.** 5
F. 6
G. 15
H. 16

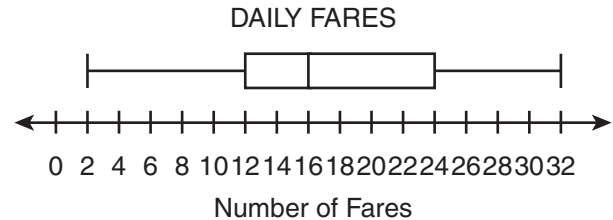
73. A company with 200 employees grew in size by 25% between 2005 and 2010. The company grew again by 10% between 2010 and 2015. What was the total number of employees in 2015?

- A.** 235
- B.** 260
- C.** 270
- D.** 275

74. A bowl contains 12 green candies, 4 yellow candies, and some red candies. The probability of choosing a green candy at random is twice as great as the probability of choosing a red candy at random. What is the probability of choosing a yellow candy at random?

- E.** $\frac{2}{11}$
- F.** $\frac{2}{9}$
- G.** $\frac{1}{4}$
- H.** $\frac{3}{11}$

75. A taxi driver recorded the number of fares each day for one month. The box plot displays the data.



Based on the box plot, which statement is a correct interpretation of the data?

- A.** The data point 2 is an outlier.
- B.** The mean of the data is 16.
- C.** The interquartile range (IQR) of the data is 30.
- D.** On about 25% of the days, there were between 12 and 16 fares.

- 76.** Serena was studying the number of water stations for different lengths of races. The table below shows the data she collected.

WATER STATIONS IN RACES

Length of Race (mi)	Number of Water Stations
3	2
12	8
18	12

Which equation represents the relationship between the length of the race, x miles, and the number of water stations, y ?

- E.** $y = \frac{2}{3}x$
- F.** $y = \frac{3}{2}x$
- G.** $y = x + 6$
- H.** $y = x - 1$

- 77.** Centerville will hold a vote on whether to build a new park. The town council surveyed a random sample of 200 residents to determine whether they would be in favor of building this park. The table shows the results of the survey.

SURVEY RESULTS

Opinion	Number of Residents
Yes	142
No	38
Undecided	20

The mayor estimates that half the undecided voters will vote in favor of building the park. Based on the survey results, if the mayor is correct, how many of the 24,100 voters will be in favor of building the park?

- A.** 17,111
- B.** 18,316
- C.** 19,401
- D.** 19,521

78. $-\frac{1}{49}, -49, -\frac{1}{7}, -1$

Which number shown above is the **greatest**?

E. $-\frac{1}{49}$

F. -49

G. $-\frac{1}{7}$

H. -1

79. What decimal is equivalent to $-\frac{13}{11}$?

A. -1.18

B. $-1.\overline{118}$

C. $-1.\overline{18}$

D. $-1.\overline{81}$

80. Each loaf of bread a baker makes requires $\frac{1}{5}$ ounce of salt. How many loaves can be made if 40 ounces of salt are used?

E. 8

F. 20

G. 45

H. 200

81. What is the value of $\frac{\left(1\frac{5}{7} - 1\frac{6}{7}\right)}{\left(3\frac{4}{7} - 3\frac{6}{7}\right)}$?

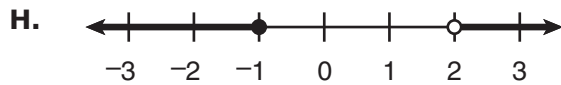
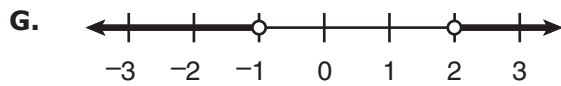
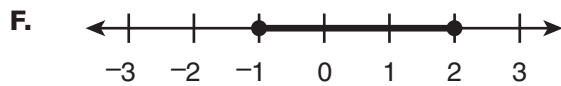
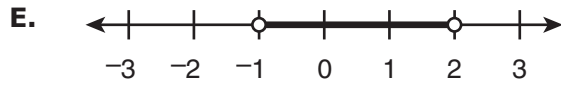
A. $-\frac{1}{2}$

B. $-\frac{1}{7}$

C. $\frac{1}{7}$

D. $\frac{1}{2}$

- 82.** Which number line represents the values of x that satisfy the inequality $-5 \leq 1 - 3x \leq 4$?



- 83.** If 1 sind = 0.75 plunk, how many sins are equivalent to 8 plunks, rounded to the nearest hundredth?

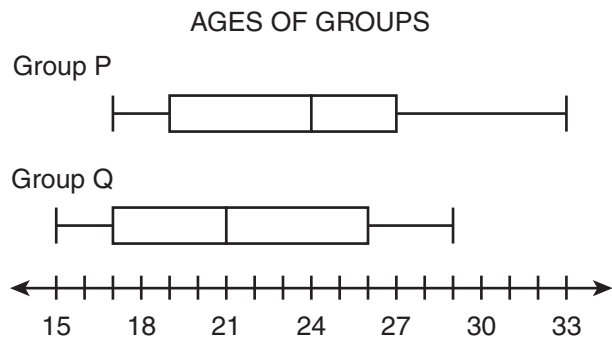
- A.** 1.33
B. 6.00
C. 7.25
D. 10.67

- 84.** Simplify:

$$8x - (7 + 2.5x) + 2$$

- E.** $5.5x - 9$
F. $5.5x - 5$
G. $10.5x - 9$
H. $10.5x - 5$

- 85.** A community center has two 30-member exercise groups. The director recorded the ages of each group's members in the box plots.



Which statement about the two groups can be verified based on the box plots?

- A.** Both groups have at least one member who is exactly 29 years old.
- B.** Group P has more members who are at least 17 years old than Group Q has.
- C.** Both groups have an equal number of members who are between 21 and 24 years old.
- D.** Group Q has fewer members who are between 19 and 27 years old than Group P has.

- 86.** Jar Q contains 12 balls, and Jar R contains 8 balls. The probability of drawing a yellow ball at random from Jar Q is $\frac{1}{3}$. The probability of drawing a yellow ball at random from Jar R is $\frac{3}{4}$. All the balls are poured into an empty jar, Jar S. What is the probability of drawing a yellow ball at random from Jar S?

- E.** $\frac{1}{4}$
- F.** $\frac{1}{3}$
- G.** $\frac{1}{2}$
- H.** $\frac{4}{7}$

87.

GIA'S MIX

Snack	Number in Bag
Pretzels	12
Raisins	9

Gia and her friends counted the numbers of pretzels and raisins in their snack mix. The table above shows the numbers in Gia's mix. Which other mix has a ratio of pretzels to raisins in the same proportional relationship as Gia's mix?

A. ADELE'S MIX

Snack	Number in Bag
Pretzels	6
Raisins	18

B. CARL'S MIX

Snack	Number in Bag
Pretzels	15
Raisins	12

C. ISABEL'S MIX

Snack	Number in Bag
Pretzels	18
Raisins	15

D. TREVON'S MIX

Snack	Number in Bag
Pretzels	16
Raisins	12

88. If $y = 4x$, what is the value of $3y + 2(3y + 5) - x$ in terms of x ?

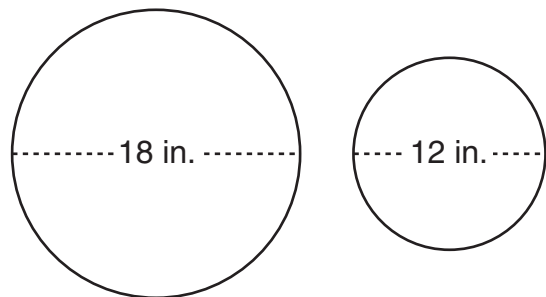
E. $8x + 10$

F. $32x + 10$

G. $35x + 10$

H. $36x + 10$

89. Two circular plates have the dimensions shown.



What is the difference between the areas of the two plates, in square inches?

A. 6π

B. 9π

C. 45π

D. 180π

90. If $x = -4$, what is the sum of x and 10% of $|x|$?

- E.** -4.4
- F.** -3.6
- G.** 0.4
- H.** 4.4

91. The volume of a cube is 512 cubic centimeters. What is the length, in centimeters, of **one** edge of this cube?

- A.** 8
- B.** $42\frac{2}{3}$
- C.** $85\frac{1}{3}$
- D.** 128

92. The high temperature on Monday was x° F. On Tuesday, the high temperature was 84° F, which was a 5% increase from Monday's high temperature. On Wednesday, the high temperature was 10% lower than it was on Monday. How much lower was the high temperature on Wednesday than the high temperature on Monday?

- E.** 4.0° F
- F.** 8.0° F
- G.** 8.4° F
- H.** 8.8° F

93. The integers $(n - 1)$, n , and $(n + 1)$ are factors of 168. What is the **greatest** possible value of n ?

- A.** 3
- B.** 7
- C.** 14
- D.** 84

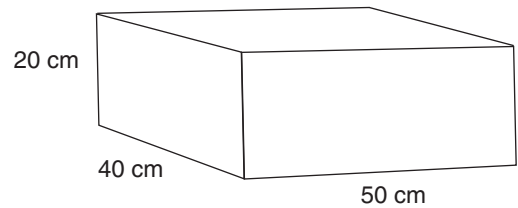
- 94.** Raoul gave two musical performances. His second performance occurred 86 days after his first performance. If Raoul's second performance was on a Tuesday, on what day of the week was his first performance?

E. Sunday
F. Monday
G. Wednesday
H. Thursday

- 95.** Michael received a check for \$213 for working 20 hours at his part-time job. The amount on the check was his total pay after \$32 in taxes was deducted. What was Michael's hourly pay rate before taxes were deducted?

A. \$9.05
B. \$10.65
C. \$12.25
D. \$21.35

96.



What is the total surface area, in square centimeters, of the closed box shown above?

E. 3,800
F. 5,600
G. 7,600
H. 40,000

- 97.** The value of a particular car decreases at a constant rate. If the car is worth \$15,000 three years after its original purchase date and \$10,000 five years after its original purchase date, what was the value of the car on its original purchase date?

A. \$18,000
B. \$20,000
C. \$22,500
D. \$25,000

-
- 98.** A bag contains 2 green, 4 blue, and 3 yellow disks. Alana selects 4 disks from the bag at random, one at a time, without replacing them. What is the probability that she selects all 4 blue disks?

E. $\frac{8}{2,187}$

F. $\frac{1}{126}$

G. $\frac{256}{6,561}$

H. $\frac{16}{9}$

-
- 99.** Jevon earns $\frac{1}{2}\%$ per year on the money in his savings account. Which expression is **not** another way to write $\frac{1}{2}\%$?

A. $\frac{0.005}{100}$

B. $\frac{\frac{1}{2}}{100}$

C. $\frac{5}{1,000}$

D. $\frac{1}{200}$

- 100.** A community garden had 50 members in 2001. In each of the next three years, the number of members increased by 40% over the number of members from the previous year. How many members were there in 2004? (Round to the nearest whole number.)

E. 70

F. 98

G. 110

H. 137

-
- 101.** Which inequality is equivalent to $6x - 20 < 5x - 2(5 - 3x)$?

A. $x < 16$

B. $x < \frac{5}{2}$

C. $x < -2$

D. $x > -2$

- 102.** A box contains a total of 30 paper clips with colors in the ratio red:white:blue = 2:5:3. Suppose that one paper clip of each color is removed from the box and not replaced. What is the probability that the next paper clip chosen will be blue?

E. $\frac{5}{27}$

F. $\frac{2}{7}$

G. $\frac{8}{27}$

H. $\frac{3}{10}$

- 103.** The least positive integer evenly divisible by the first 6 positive integers is 60. What is the least positive integer evenly divisible by the first 8 positive integers?

- A.** 1,680
B. 840
C. 480
D. 420

- 104.** Three different colors of paint are available to paint the window frame, the door, and the wall of a house. If no two parts of the house may be painted the same color, how many different ways are there to paint the three parts of the house?

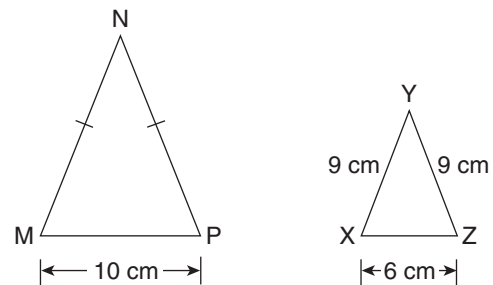
E. 3

F. 6

G. 9

H. 27

105.



Triangle MNP is similar to triangle XYZ, and $MN = NP$. What is the perimeter of triangle MNP?

- A.** 25 cm
B. 30 cm
C. 36 cm
D. 40 cm

106. Suppose that w , x , y , and z are integers, none of which are zero. Under which set of conditions would the product $x^2yz^3|w|$ **always** be positive?

- E.** x is positive and w is positive.
- F.** x is negative and w is negative.
- G.** y is negative and z is negative.
- H.** x and y are negative, and z and w are positive.

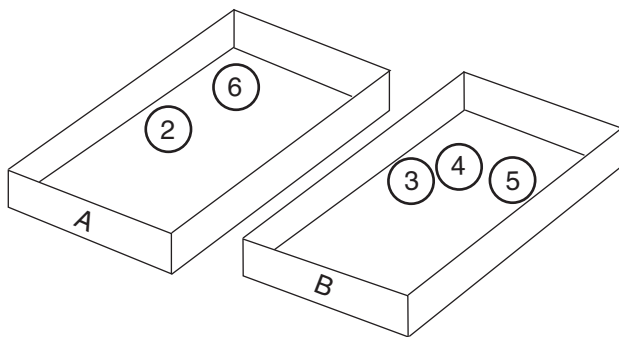
107. Chris has 1 red shirt, 2 white shirts, and 3 black shirts. Chris picks a shirt at random. Without putting the first shirt back, he picks a second shirt at random. What is the probability that the first shirt is black and the second shirt is red?

- A.** $\frac{1}{12}$
- B.** $\frac{1}{10}$
- C.** $\frac{1}{5}$
- D.** $\frac{7}{10}$

108. Rita used 125 small solid cubes to build a large solid cube. The measure of each edge of the small cubes equals 1 inch. What is the measure of each edge of the large cube, in inches?

- E.** 5
- F.** 25
- G.** 41.33
- H.** 125

109.



Five numbered balls are placed in two boxes as shown in the figure above. Davina picks one ball at random from Box B and places it in Box A. Then Yusuf picks one ball from Box A without looking. What is the probability that Yusuf will pick a ball with an odd number?

- A. $\frac{2}{9}$
- B. $\frac{1}{3}$
- C. $\frac{2}{5}$
- D. $\frac{2}{3}$

110. Which graph represents the solution to

$$|x + 1| \geq 2?$$

- E.
- F.
- G.
- H.

111. A rectangular kitchen floor is 12 feet long and 7.5 feet wide. It will be covered completely with new square tiles measuring 0.75 foot on each side. How many tiles will cover the floor?

- A. 68
- B. 78
- C. 120
- D. 160

112. Typically, an amusement park sells 3 times as many tickets on Sunday as on any weekday, and 4 times as many tickets on Saturday as on Sunday. If the number of tickets sold on Saturday and Sunday totaled 30,000, how many tickets are expected to be sold on the following Monday?

- E.** 2,000
- F.** 2,500
- G.** 3,000
- H.** 4,285

113. A rectangular field had a length of 80 meters and a width of 50 meters. The field was recently enlarged, with the length increased by 15% and the width increased by 10 meters. By what percentage was the area of the field increased?

- A.** 26.5%
- B.** 35%
- C.** 38%
- D.** 42.5%

114. If x and y are both negative integers with $x < y$, which expression has the **least** value?

- E.** x^2y
- F.** xy^2
- G.** $-x^2y$
- H.** $-xy^2$

THIS IS THE END OF THE TEST.
IF TIME REMAINS, YOU SHOULD CHECK
YOUR ANSWERS. BE SURE THAT THERE
ARE NO STRAY MARKS, PARTIALLY
FILLED ANSWER CIRCLES, OR
INCOMPLETE ERASURES ON YOUR
ANSWER SHEET. ■